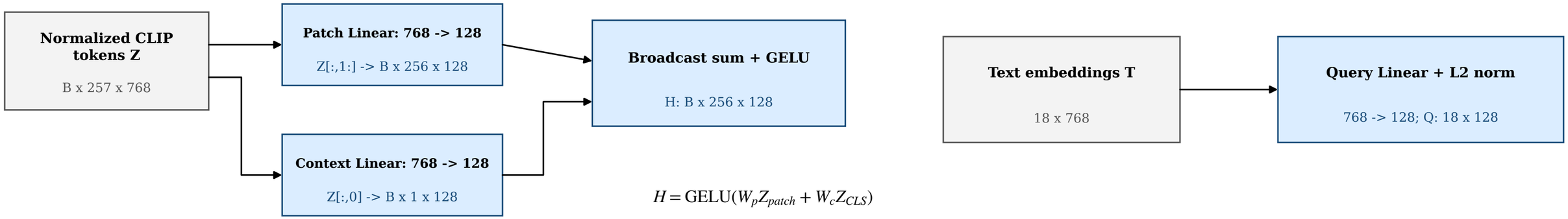
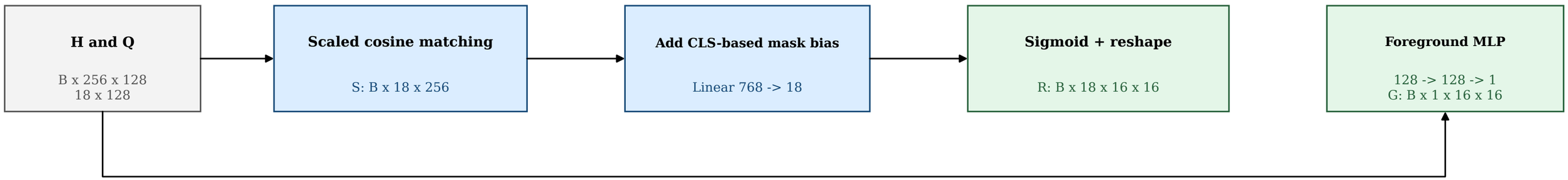
Inside the Learnable Concept & Region Predictor

(a) Build image patch features and text queries



$$H = \mathrm{GELU}(W_p Z_{patch} + W_c Z_{CLS})$$

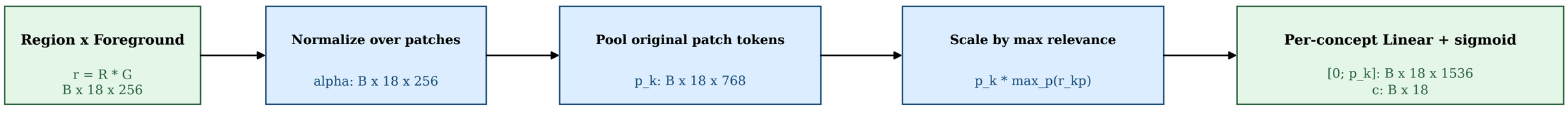
(b) Predict foreground G and all 18 concept regions R



$$S_{bkp} = s\,\langle \mathrm{norm}(H_{bp}), Q_k \rangle; \quad R = \mathrm{sigmoid}(S + W_m Z_{CLS})$$

Foreground: Linear -> GELU -> Linear -> sigmoid

(c) Use the predicted regions to read native concept probabilities



$$\alpha_{bkp} = r_{bkp} / \sum_p r_{bkp}$$

$$p_{bk} = \sum_p \alpha_{bkp} Z_{bp}$$

Readout weights: 18 x 1536; the first 768 inputs are zero in the region-readout model.